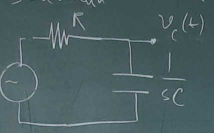
PS2

Last class (...) : Given $v_p(t) = K\sin(\omega t + \phi)$ & solve for K, ϕ

We go for Route 3 (recap of RLC)

s-domain



$V \cdot \frac{\omega}{s^2+\omega^2}$

Φ_L

$V\sin(\omega t)$

$v_c(0)=0$

$$V_c(s) = \boxed{\frac{\omega}{s^2+\omega^2}} \cdot \frac{1}{R+\frac{1}{sC}} \cdot \frac{1}{sC} = \frac{\omega}{s^2+\omega^2} \cdot \frac{1}{s+\frac{1}{RC}} \frac{1}{RC}$$

$$= \frac{1}{RC} \frac{\omega}{(s+j\omega)(s-j\omega)} \cdot \frac{1}{s+\frac{1}{}} = \frac{A_1}{s+j\omega} + \frac{A_2}{s-j\omega} + \frac{A_3}{s+}$$

$$A_1 = \frac{1}{-2j\omega} \cdot \frac{1}{-j\omega+\frac{1}{RC}} \quad \boxed{A_1 = A_2^*}$$

$A_2 = \frac{1}{}\ \frac{1}{}$

$$\underbrace{a\frac{dv_p}{dt} + bv_p} \rightarrow A_1\cos(\omega_p t+\phi) + A_2\sin(\omega_p t+\phi)$$

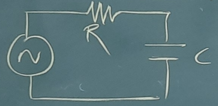
$\Downarrow$ = Sine

Route 2 → Given is correct! just solve for K, ϕ

Recap

Goal

$u(t) \cdot V\sin(\omega t)$



1st: $v_{in}(t) = f(t)$

$$\left[\frac{e^{j\omega t}-e^{-j\omega t}}{2j}\right]$$

2-exp   superposition

Route 1

PS1

initial conds were part of "Laplace"

We did using Laplace

$f(t) = V u(t)$

or

→ lin comb of $u(t)$

pblx

Final soln.

$v_p(t) + A \cdot v_N(t)$

initial cond will give it!

$a\frac{dv}{dt} + bv = f(t)$

particular soln (forced response)

if $f(t) = e^{\alpha t}$

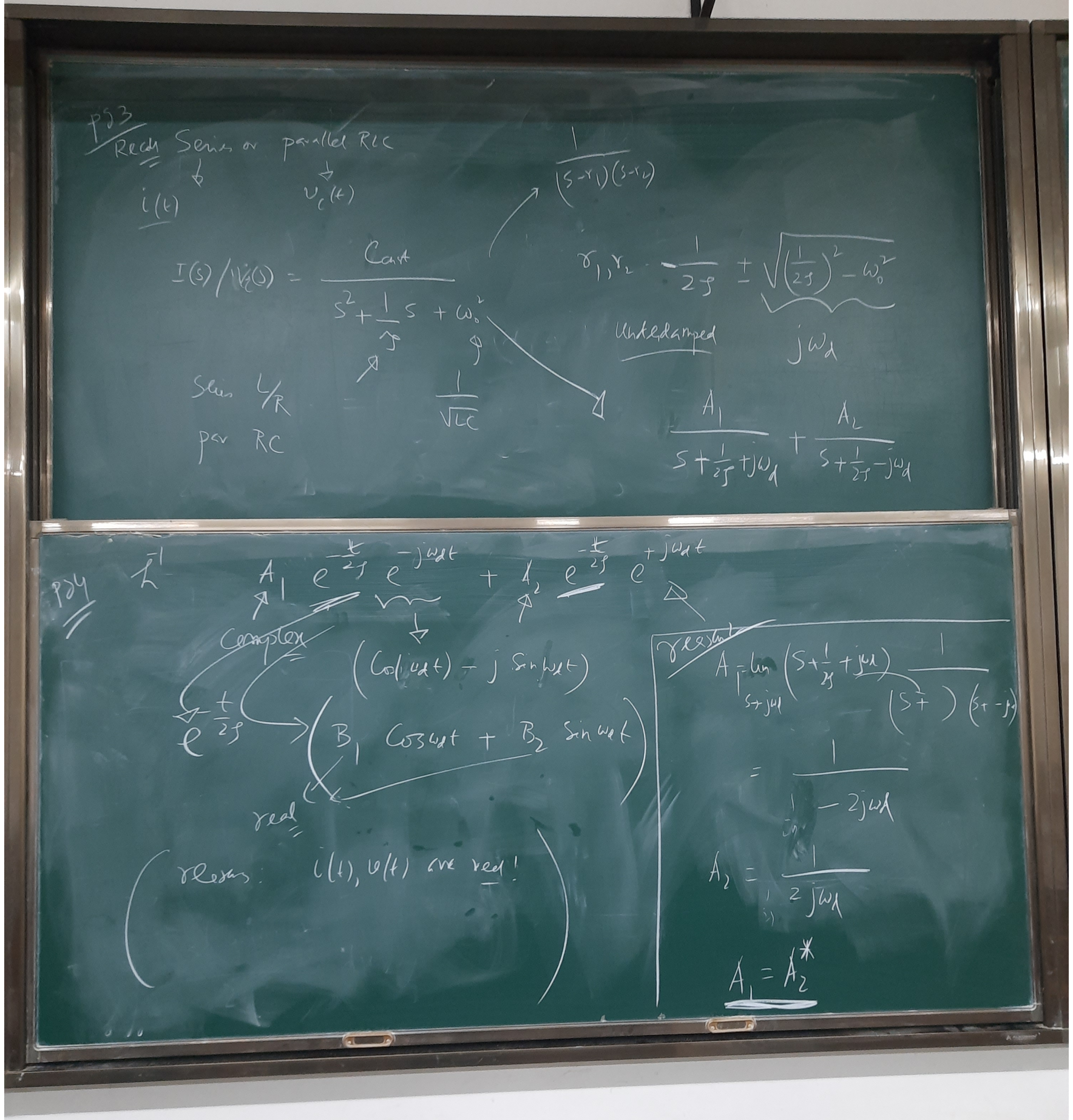
then $v_p(t) = Ke^{\alpha t}$

get K by solving

$v_p(t)$ ← Given & check that $a\frac{dv_p}{dt} + bv_p = f(t)$

transient/natural: $a\frac{dv}{dt} + bv = 0 \Rightarrow v_N(t)$

$a\frac{d^2v}{dt^2} + b\frac{dv}{dt} + cv = 0 \Rightarrow$ { two complex roots



p23

Recall Series or parallel RLC

$i(t)$ $\qquad v_c(t)$

$$I(s)/V_c(s) = \frac{\text{Cont}}{s^2 + \frac{1}{\rho}s + \omega_0^2} \rightarrow \frac{1}{(s-r_1)(s-r_2)}$$

Series L/R $\qquad \frac{1}{\sqrt{LC}}$

par RC

$$r_1, r_2 = -\frac{1}{2\rho} \pm \underbrace{\sqrt{\left(\frac{1}{2\rho}\right)^2 - \omega_0^2}}_{j\omega_d}$$

Underdamped

$$\frac{A_1}{s+\frac{1}{2\rho}+j\omega_d} + \frac{A_2}{s+\frac{1}{2\rho}-j\omega_d}$$

p24

$$\mathcal{L}^{-1} \quad A_1 e^{-\frac{t}{2\rho}} e^{-j\omega_d t} + A_2 e^{-\frac{t}{2\rho}} e^{+j\omega_d t}$$

Complex

$$(\cos(\omega_d t) - j\sin\omega_d t)$$

$$e^{-\frac{t}{2\rho}} \left(B_1 \cos\omega_d t + B_2 \sin\omega_d t\right)$$

real

(reason: $i(t)$, $v(t)$ are real!)

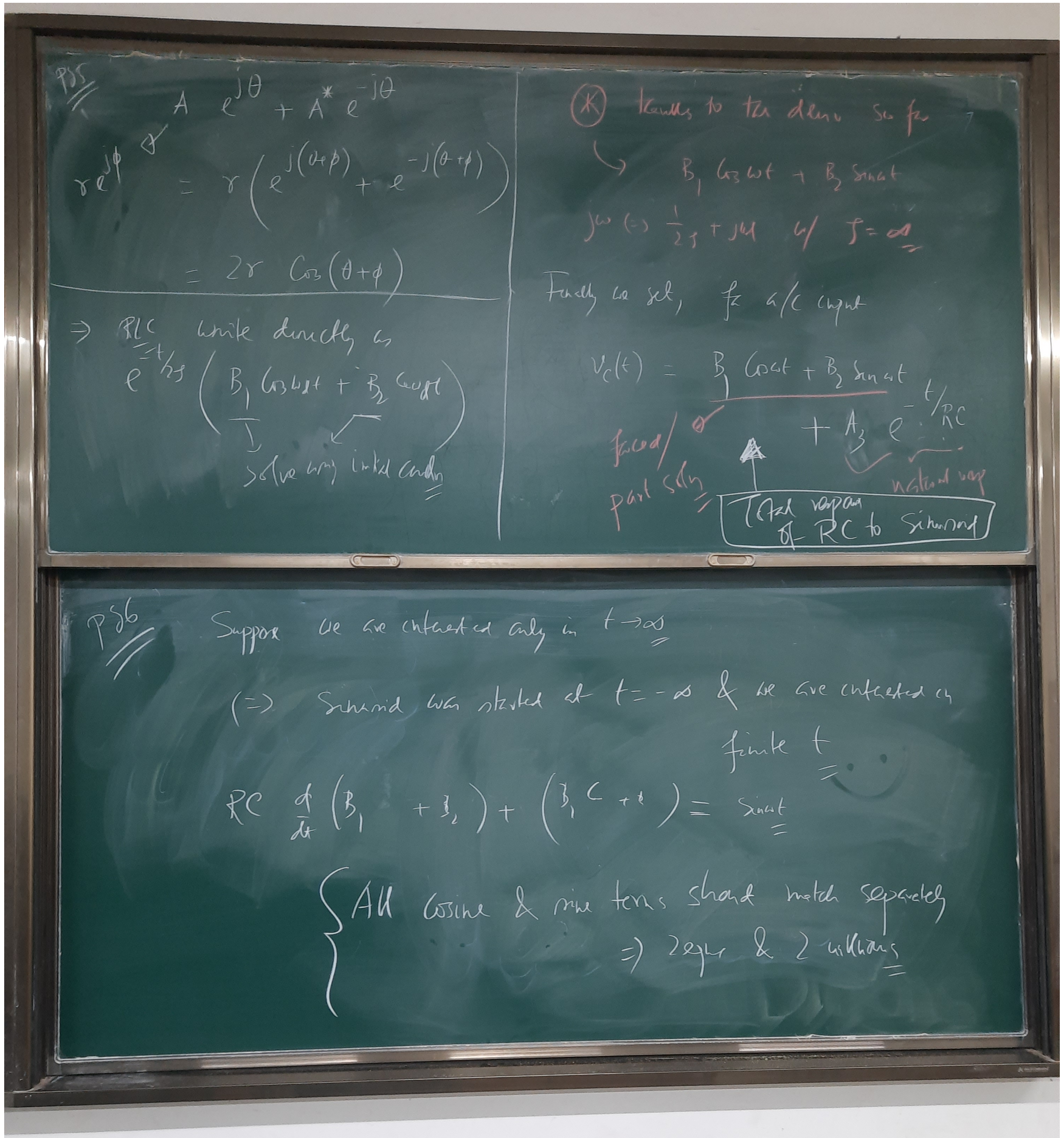
result

$$A_1 = \lim_{s \to -j\omega_d} \left(s+\frac{1}{2\rho}+j\omega_d\right) \frac{1}{(s+\ )(s+\ -j\omega_d)}$$

$$= \frac{1}{-2j\omega_d}$$

$$A_2 = \frac{1}{2j\omega_d}$$

$$A_1 = A_2^*$$



PS5

$$A e^{j\theta} + A^{*} e^{-j\theta}$$

$$r e^{j\phi}$$

$$= r\left(e^{j(\theta+\phi)} + e^{-j(\theta+\phi)}\right)$$

$$= 2r\cos(\theta+\phi)$$

⇒ RLC write directly as

$$e^{-t/2\zeta}\left(B_1\cos\omega_d t + B_2\cos\omega_d t\right)$$

solve using initial condn

Ⓧ Recalls to the demo so far

↳ $B_1\cos\omega t + B_2\sin\omega t$

$j\omega \Leftrightarrow \frac{1}{2\zeta} + j\omega$ w/ $\zeta = \infty$

Finally we get, for a/c input

$$V_C(t) = \underbrace{B_1\cos\omega t + B_2\sin\omega t}_{\text{forced/part soln}} + \underbrace{A_3 e^{-t/RC}}_{\text{natural resp}}$$

$\boxed{\text{Total response of RC to sinusoid}}$

PS6

Suppose we are interested only in $t\to\infty$

(⇒ Sinusoid was started at $t=-\infty$ & we are interested in finite t

$$RC\frac{d}{dt}\left(B_1 \quad + B_2 \quad\right) + \left(B_1 C + \quad\right) = \sin\omega t$$

All cosine & sine terms should match separately
⇒ 2 eqns & 2 unknowns